CUBICAL MODELS WITH REVERSAL DO NOT PRESENT SPACES: THE INTERVAL QUOTIENT IS A $K(\mathbb{Z}/2, 1)$

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Abstract. Since 2018 it has been folklore, recorded only in a lecture and a mailing-list discussion, that the cubical-type model structure on De Morgan cubical sets — the semantic home of CCHM cubical type theory — does not present the homotopy theory of spaces: the quotient L of the representable interval by the reversal $x \mapsto \neg x$ has contractible geometric realization but is not weakly contractible. We give the first complete proof, in a form that covers the De Morgan, Kleene and Boolean cube categories uniformly — the Kleene case having never been examined — and we identify the homotopy type: L is a $K(\mathbb{Z}/2, 1)$ in the cubical-type model structure. The proof has two engines. A one-line *freeness theorem* shows the reversal acts without fixed cells on the interval, in every dimension and all three theories; a finite *reachability census* of reversal-symmetric squares then closes the non-contractibility argument, adapting Coquand's trivial-cofibration/fibration factorization method from the cartesian symmetric square to the symmetric line — including two subtleties absent from the cartesian case (connection-degenerate composition directions, and symmetric contractibility of the connection-degenerate line, which makes the reachability census essential). Freeness then powers a *sheet-transport* fibration structure on the covering $\square^1 \to L$, whose total space is contractible by a connection and whose fiber is the discrete $\mathbb{Z}/2$: the quotient computes the homotopy quotient, not the orbit space. Consequences: the cubical-type and test model structures differ on all three cube categories, and geometric realization is not a Quillen equivalence for any of them.

1. Introduction

Cubical type theory in the style of Cohen–Coquand–Huber–Mörtberg [5] is interpreted in presheaves on a cube category whose interval carries connections $\wedge, \vee$ and a reversal $\neg$ — the free De Morgan algebra structure. These presheaves carry a *cubical-type model structure* (Sattler [10]; Coquand [6] gives an explicit trivial-cofibration/fibration factorization), whose fibrant objects model univalent type theory. Whether this model structure presents

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the homotopy theory of spaces was answered in the negative, for the cartesian cube category, by an argument of Sattler and Coquand in June 2018: the quotient of the square by the coordinate swap is not weakly contractible, though its realization is a triangle [6, 7]. The equivariant repair of this defect for cartesian cubes is the recent landmark of Awodey–Cavallo–Coquand–Riehl–Sattler [1].

For the De Morgan cube category the corresponding negative claim — with counterexample the quotient

$$L \;:=\; \square^1/(x \sim \neg x)$$

of the representable interval by the reversal — has circulated since the same discussions [7, 9], but no proof has been published: Coquand's note proves the cartesian case in full and says of the De Morgan case only that the argument "should also be valid" [6, Rem. 3]; the recent literature [1, 4] continues to cite the 2018 lecture [9] for it. The Kleene cube category — the canonical cube category of the standard interval $[0, 1]$ [2] — appears in this connection only once in the literature, as a parenthetical remark in the 2018 thread [7].

This paper closes the question, in a stronger form than the folklore records. Throughout, $\mathcal{T}$ denotes any of the De Morgan, Kleene, or Boolean cube categories, and "the model structure" means the cubical-type model structure on presheaves over $\mathcal{T}$.

Theorem A (= Theorem 4.5)**.** In all three theories, $L \to 1$ is not a weak equivalence, although the geometric realization $|L| \cong [0, 1/2]$ is contractible. Hence the cubical-type model structure differs from the test model structure, and geometric realization is not a Quillen equivalence.

Theorem B (= Theorem 5.2)**.** In all three theories, the quotient map $p : \square^1 \to L$ is a fibration with contractible total space and fiber the discrete $\mathbb{Z}/2$; consequently L is a $K(\mathbb{Z}/2, 1)$ in the cubical-type model structure.

Theorem A for De Morgan and Boolean cubes is the first published proof of the folklore claim; for Kleene cubes it is new. Theorem B identifies the actual homotopy type, which no source has claimed: the moral is that *the presheaf quotient computes the homotopy quotient of the reversal action, not its orbit space* — exactly the phenomenon that equivariance was invented to control on the cartesian side [1], here appearing for the $\mathbb{Z}/2$-action of the reversal, for which no equivariant theory yet exists.
Method. Everything flows from one observation (Theorem 3.1): *the reversal acts freely on the cells of the interval* — no reparametrization t of the interval satisfies $t = \neg t$, in any dimension and any of the three theories, because the all-consistent vertex assignment is fixed by the point-level involution while Boolean values are not. Freeness has two independent uses. For Theorem A we adapt Coquand's invariant method [6, §4–5] from symmetric squares to *symmetric lines*; two subtleties absent from the cartesian case appear. First,

the composition directions fixed by the reversal now include the connection-degenerate elements $x \wedge \neg x$ and $x \vee \neg x$, enlarging the case analysis of the factorization argument. Second — and this is where the cartesian intuition genuinely fails — the connection-degenerate line $\ell(x \wedge \neg x)$ *is* symmetrically contractible, so the chain version of the invariant could conceivably connect the generator to a constant through it; whether it can is a finite question about two-variable free algebras, and an exhaustive *reachability census* answers it: the symmetric-square graph on symmetric lines decomposes, in all three theories, into the isolated component of the generator and the contractible component of the degenerate lines. (The census sizes are small: 60, 32 and 8 symmetric squares for De Morgan, Kleene, Boolean.) For Theorem B, freeness makes the two lifts of any cell of L into disjoint *sheets*, and the fibration structure on p is *sheet transport*: the unique lift through a prescribed face, with a tube-forcing lemma showing all compatibility data lies on the same sheet. The reachability census and the monodromy of the covering tell one story: the generator line has odd parity, the degenerate lines even — the combinatorial shadow of $\pi_1 = \mathbb{Z}/2$.
Relation to the literature. Our debt to [6] is total as to method; the contribution is the adaptation (with its two new subtleties), the uniform treatment of the three theories, the previously unexamined Kleene case, the identification of the homotopy type, and machine verification of every finite ingredient. The reversal-elimination theorem of Cavallo–Sattler [4] obtains a model of a theory *with* reversals presenting spaces by interpreting over reversal-free cubes; Theorems A and B delimit what it is they are eliminating: reversal as a *cube-category* symmetry inevitably manufactures $K(\mathbb{Z}/2,1)$s from free quotients. Whether an equivariant fibration structure for the reversal action — the $\mathbb{Z}/2$, or in higher dimensions hyperoctahedral, analogue of [1] — repairs the De Morgan and Kleene models is, we believe, the natural open problem these results isolate (§7). The literal-cube combinatorics of free De Morgan and Kleene algebras used throughout is developed in [12, 13].

2. Preliminaries

2.1. **Cube categories and the interval.** Following [6, §1], the base category has as objects finite sets of names $I, J, \ldots$; a morphism $f : J \to I$ is a function from I to the free algebra $\mathcal{T}(J)$ of the interval theory $\mathcal{T}$ on the names J, where $\mathcal{T}(J)$ is the free De Morgan algebra $\mathrm{DM}(J)$, the free Kleene algebra $\mathrm{KL}(J)$, or the free Boolean algebra $\mathrm{BA}(J)$. The formal interval is $\mathbb{I}(J) = \mathcal{T}(J)$. We use the literal-cube representations [12, 13]: $\mathrm{DM}(J)$ is the lattice of monotone Boolean functions on $\{0,1\}^{2|J|}$ (coordinates in pairs $(x, \neg x)$), $\mathrm{KL}(J)$ its quotient by the mixed points, $\mathrm{BA}(J)$ the functions on consistent points. The reversal $\neg$ acts on elements by $(\neg t)(\alpha) = 1 - t(\sigma\bar{\alpha})$, where $\bar{\alpha}$ complements all coordinates and σ swaps each pair.

$\square^1 = \mathrm{Yon}(x)$ denotes the representable interval, with cells $\square^1(J) = \mathcal{T}(J)$, and

$$L \;=\; \square^1/(t \sim \neg t), \qquad L(J) = \mathcal{T}(J)/{\sim},$$

the quotient presheaf (colimits are computed pointwise). We write $[t]$ for cells of L and $\ell(t)$ when we think of them as reparametrizations of the generating line. Note $[0] = [1]$: L has a single vertex ℓ_0, and the generator $[x]$ is a loop at it.

2.2. **The model structure and the factorization.** The cubical-type model structure on $\widehat{\mathcal{T}}$ has cofibrations the (decidable) monomorphisms; its fibrations are defined by a uniform composition operation $\mathsf{comp}^{k\to l}$ against partial elements, and Coquand's note constructs, for any map $\sigma : A \to B$, an explicit trivial cofibration–fibration factorization $A \to T \to B$ where T is the total space of an inductively defined family E over B with constructors inc and $\mathsf{comp}^{k\to l}(u_k, \psi, u, v)$ ($k \neq l \in \mathbb{I}(J)$, $\psi \neq 1$), and restriction maps given by: reindexing when $kg \neq lg$ and $\psi g \neq 1$; $u_k g$ when $kg = lg$; $u_{[l]g}$ when $\psi g = 1$ [6, §3]. A map is a weak equivalence iff in (any) such factorization the fibration part is a trivial fibration; trivial fibrations have sections. All of this is verbatim uniform in $\mathcal{T}$: the constructions never inspect the interval theory beyond its algebra operations.

3. The freeness theorem

Theorem 3.1 (freeness)**.** *In all three theories and for every J, no $t \in \mathcal{T}(J)$ satisfies $t = \neg t$. Hence the reversal action on the cells of $\square^1$ is free, and every cell of L has exactly two lifts $\{t, \neg t\}$.*

Proof. Let α_0 be the point of the literal cube with every pair equal to $(0,1)$ — the consistent point of the all-zeros assignment, present in all three representations. Then $\sigma\bar{\alpha}_0 = \alpha_0$: complementing gives all pairs $(1,0)$, and swapping each pair returns $(0,1)$. So $t = \neg t$ would give $t(\alpha_0) = 1 - t(\sigma\bar{\alpha}_0) = 1 - t(\alpha_0)$, impossible for a Boolean value. $\square$

(Machine census: there are 0 self-dual elements among all 7,828,354 elements of DM(3), all 43,918 of KL(3), and below; see §6.)

4. Theorem A: L is not weakly contractible

Definition 4.1. A square $c(x,z)$ *symmetrically connects* lines $a(x)$ and $b(x)$ if $c(x,0) = a$, $c(x,1) = b$ (or vice versa) and $c(\neg x, z) = c(x,z)$. Write $P'(A)$ for the property: every reversal-symmetric line $a(x) = a(\neg x)$ of A is connected to a constant line by a finite chain of symmetrically connecting squares. (In a quotient like L, symmetry of $[t]$ means $t \circ \sigma_x \in \{t, \neg t\}$.)

Lemma 4.2. *If $u : A \to B$ has a section then $P'(A)$ implies $P'(B)$.*

Proof. As in [6, Lem. 4.1]: push the symmetric line along the section, connect there, map the chain back. $\square$

Proposition 4.3. *For every $\sigma : A \to B$ with factorization $A \to T \to B$ as above, $P'(A)$ implies $P'(T)$.*

Proof. A symmetric line of T over $v[l]$ is either $(\sigma a, \mathsf{inc}\, a)$ with a a symmetric line of A — handled by $P'(A)$ — or $(v(x,l), \mathsf{comp}^{k\to l}(a, \psi, u, \langle z\rangle v(x,z)))$ with $k \neq l \in \mathbb{I}(x)$. Two remarks before the induction. First, the substitution $g = (x \mapsto \neg x)$ is an *automorphism* of $\mathcal{T}(J)$, so the degenerate restriction clauses cannot fire under it: $kg = lg$ iff $k = l$, and $\psi g = 1$ iff $\psi = 1$. Symmetry of the cell is therefore an equality of comp-constructors, forcing $k \circ \sigma_x = k$, $l \circ \sigma_x = l$, a symmetric, ψ, u, v symmetric. Second, the fixed elements of $\mathbb{I}(x)$ under $x \mapsto \neg x$ are $\{0, 1\}$ in the Boolean theory but

$$\{\, 0,\ 1,\ x \wedge \neg x,\ x \vee \neg x \,\}$$

in the De Morgan and Kleene theories: the composition may enter or exit through a connection-degenerate direction, a case with no cartesian analogue. The induction is nevertheless uniform: with a fresh name t, the square

$$c(x,t) \ = \ \big(v(x,t),\ \mathsf{comp}^{k\to t}(a, \psi, u, \langle z\rangle v(x,z))\big)$$

is symmetric (all its data is), equals the given line at $t = l$, and at $t = k$ equals $(v[k], a)$ by the $kg = lg$ restriction clause. Since a has smaller inductive height and is symmetric, induction reaches an inc-cell and $P'(A)$ concludes. □

Proposition 4.4. *$P'(L)$ fails, in all three theories.*

Proof. The symmetric lines of L are exactly its three (Boolean: two) lines: the generator $[x]$, the connection-degenerate $[x \wedge \neg x] = [x \vee \neg x]$ (absent in the Boolean case), and the constant $[0] = [1]$: every element of $\mathcal{T}(x)$ is orbit-symmetric. A symmetrically connecting square of L is an orbit $[t]$, $t \in \mathcal{T}(x,z)$, with $t \circ \sigma_x \in \{t, \neg t\}$ — a *finite* set, and the faces $z = 0, 1$ of its members are computable. The exhaustive census (§6): the symmetric squares number 60 (De Morgan), 32 (Kleene), 8 (Boolean), and the induced graph on symmetric lines is, in each theory,

$$\underbrace{[x] \circlearrowleft}_{\text{isolated}} \qquad\qquad [x \wedge \neg x] \longleftrightarrow \text{const} \ \circlearrowleft,$$

i.e. every symmetric square with one face $[x]$ has both faces $[x]$. Two features deserve note. The degenerate line *is* symmetrically contractible — $t = (x \wedge \neg x) \wedge \neg z$ is exactly symmetric and connects it to 0 — so the isolation of $[x]$ is essential and not a foregone conclusion; and the connection trick that kills the cartesian symmetric-square invariant [6, Rem. 1] produces $\ell(x \vee z)$, which is *not* symmetric ($x \vee z \neq \neg x \vee z$ and $\neq \neg(x \vee z)$ in the free algebras), in accordance with the census. Hence no chain connects $[x]$ to a constant. □

Theorem 4.5. *In all three theories $L \to 1$ is not a weak equivalence; the cubical-type model structure differs from the test model structure; and geometric realization is not a Quillen equivalence.*

Proof. If $L \to 1$ were a weak equivalence then so would be the vertex $1 \to L$ (two-out-of-three); factor it as $1 \to T \to L$; the fibration $T \to L$ is then trivial, hence has a section. $P'(1)$ holds trivially, Proposition 4.3 gives $P'(T)$, and Lemma 4.2 gives $P'(L)$ — contradicting Proposition 4.4. For the comparisons: $|L| \cong [0,1]/(s \sim 1-s) \cong [0,1/2]$ is contractible, and all three cube categories are strict test categories [2], so in the test model structure $L \to 1$ *is* a weak equivalence — the two localizations differ. Since cofibrations are monomorphisms, L is cofibrant, and a left Quillen equivalence reflects weak contractibility of cofibrant objects [8, 1.3.16]; realization fails to do so. $\square$

5. Theorem B: L is a $K(\mathbb{Z}/2, 1)$

Theorem 5.1 (sheet transport)**.** *The quotient $p : \square^1 \to L$ is a fibration: it carries a uniform composition structure.*

Proof. In dependent form, the fiber over a cell $v \in L(J)$ is $E(v) = \{t, \neg t\}$, the two lifts (Theorem 3.1). Given composition data — $v \in L(J^+)$, $k \neq l \in \mathbb{I}(J)$, a lift $u_k \in E(v[k])$, $\psi \neq 1$, and a compatible partial family $u_f \in E(vf)$ for $f : K \to J^+$ with $\psi \mathsf{p} f = 1$ — there is a *unique* lift $T \in \mathcal{T}(J^+)$ of v with $T[k] = u_k$: of the two lifts $T, \neg T$ exactly one restricts to u_k, since $T[k] = \neg T[k]$ would contradict freeness at level J. Define

$$\mathsf{comp}^{k \to l}(u_k, \psi, u) \ := \ T[l].$$

Tube forcing: for $g : K \to J$ with $\psi g = 1$, the tube cell u_{g^+} (part of the data, as $\psi \mathsf{p} g^+ = 1$) satisfies $u_{g^+}[kg] = u_{[k]g} = u_k g = T[k]g = (Tg^+)[kg]$, using $[k] \circ g = g^+ \circ [kg]$; by freeness at level K^+, agreeing with Tg^+ at one face forces $u_{g^+} = Tg^+$, whence $u_{[l]g} = u_{g^+}[lg] = (Tg^+)[lg] = T[l]g$: the composition agrees with the prescribed data on ψ. Uniformity: for g with $kg \neq lg$, $\psi g \neq 1$, the unique lift of vg^+ through $u_k g$ is Tg^+, so $\mathsf{comp}(\dots)g = T[l]g = \mathsf{comp}(u_k g, \psi g, ug^+)$ — naturality of the construction. Finally $\mathsf{comp}^{k \to k} = T[k] = u_k$. $\square$

Theorem 5.2. *In all three theories, L is a $K(\mathbb{Z}/2, 1)$: the loop object of its fibrant replacement at the vertex is weakly equivalent to the discrete two-point object, and its higher homotopy vanishes.*

Proof. p is a fibration (Theorem 5.1) with fiber over the vertex the discrete two-point presheaf. Its total space $\square^1$ is contractible: the connection square $c(x, z) = x \wedge \neg z$ is a homotopy from the identity to the constant 0. The fiber sequence $\mathbb{Z}/2 \to \square^1 \to L$ then identifies, in the homotopy category of the model structure, the loop object of L at ℓ_0 with the discrete fiber and kills the higher homotopy: $\pi_n(L) = \pi_{n-1}(\mathbb{Z}/2) = 0$ for $n \geq 2$. Concretely, the generator loop $[x]$ lifts to the path x from 0 to 1, crossing sheets: its monodromy is the nontrivial element of $\mathbb{Z}/2$ (this re-proves Theorem 4.5 independently of §4), while $[x] \cdot [x]$ lifts to a genuine loop and is null-homotopic — $\pi_1 = \mathbb{Z}/2$ exactly. $\square$

Remark 5.3. The two proofs of non-contractibility are one: the reachability census of Proposition 4.4 separates the symmetric lines precisely by the parity of their monodromy — $[x]$ lifts $0 \to 1$ (odd), while $[x \wedge \neg x]$ and the constants lift to loops (even) — and symmetric squares preserve the parity. The census is the combinatorial shadow of $\pi_1 = \mathbb{Z}/2$.

Remark 5.4. Theorem 5.2 says the presheaf quotient of the interval by the reversal computes the *homotopy* quotient $(\square^1)_{h\mathbb{Z}/2} \simeq B\mathbb{Z}/2$ rather than the orbit space — because the action is free on cells (Theorem 3.1) even though its topological realization has a fixed point at $\frac{1}{2}$. This is precisely the mechanism that equivariant fibrations neutralize on the cartesian side [1], where the symmetric quotients I^n/H play the role of L; the reversal case is, if anything, cleaner, since freeness makes the covering honest. No equivariant theory for the reversal ($\mathbb{Z}/2$ per direction; hyperoctahedrally, $\Sigma_n \ltimes (\mathbb{Z}/2)^n$ in dimension n) exists in the literature.

6. Machine verification

All finite ingredients were verified exhaustively (scripts accompany the artifact). *Freeness*: 0 self-dual elements in $\mathrm{DM}(m)$ for $m \leq 3$ (out of 6, 168, 7,828,354) and in $\mathrm{KL}(m)$ for $m \leq 3$ (out of 6, 84, 43,918) — the theorem's one-line proof makes higher m unnecessary, but the census independently guards the representations. *No rescue terms*: no t in $\mathrm{DM}(x,z)$ or $\mathrm{KL}(x,z)$ is reversal-compatible, restricts to the generator at $z = 0$, and is constant at $z = 1$ — the connection trick of [6, Rem. 1] has no symmetric analogue. *Reachability*: the symmetric squares number 60 / 32 / 8 (De Morgan / Kleene / Boolean); their face graphs on symmetric lines have edge counts $[x] \to [x]$: 10/6/4, degenerate $\leftrightarrow$ constant: 10+10 / 6+6 / — , constant $\to$ constant: 10/8/4, and *no* edge leaves the $[x]$-component in any theory. *L-cells*: the squares of L over the generator number 62 (44 with z-varying faces, 12 contracting the generator non-symmetrically) — the quotient is rich; only symmetry starves it.

7. Outlook

Three directions. (1) *Reversal equivariance*: define uniform fibrations equivariant for the $\mathbb{Z}/2$-actions of reversal (hyperoctahedrally, $\Sigma_n \ltimes (\mathbb{Z}/2)^n$) on De Morgan or Kleene cubes, in the style of [1]; the freeness theorem says the reversal orbits are regular on the interval, so the equivariant filling conditions have no isotropy at the crucial cells — we regard the repair as plausible and the natural continuation. (2) *The remaining open positive case*: our argument uses reversal essentially; for Dedekind cubes (no reversal) the presentation question remains open [11, 3], with a constructive solution announced by Sattler. (3) *Stronger inequivalence*: Theorems A and B show the canonical comparison fails; whether the cubical-type localization is inequivalent to spaces under *every* functor — asserted in the folklore, and explicitly not excluded on the nLab — would follow from an invariant distinguishing the localizations; the abundance of $K(\mathbb{Z}/2,1)$s with contractible realization produced here (one for every free reversal quotient) is a start.

Artifact. The verification scripts and the machine-checked companions are available in the artifact repository:
`https://github.com/r-shibusawa/cubical-strict-j`, archived at DOI 10.5281/zenodo.21405961.

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